

How Do People Use a Public Gratitude Platform in the Wild?

Min Zhang

The Open University, Walton Hall,
Milton Keynes, UK, MK7 6AA
min.zhang@open.ac.uk

Carlos Arce-Plata

The Open University, Walton Hall,
Milton Keynes, UK, MK7 6AA
carlos.arce-plata@open.ac.uk

Blaine Price

The Open University, Walton Hall,
Milton Keynes, UK, MK7 6AA
b.a.price@open.ac.uk

Graham Pike

The Open University, Walton Hall,
Milton Keynes, UK, MK7 6AA
graham.pike@open.ac.uk

Zoe Walkington

The Open University, Walton Hall,
Milton Keynes, UK, MK7 6AA
z.walkington@open.ac.uk

Camilla Elphick

The Open University, Walton Hall,
Milton Keynes, UK, MK7 6AA
camilla.elphick@open.ac.uk

Lara Frumkin

The Open University, Walton Hall,
Milton Keynes, UK, MK7 6AA
lara.frumkin@open.ac.uk

Richard Philpot

Lancaster University, Bailrigg,
Lancaster, UK, LA1 4YW
r.philpot@lancaster.ac.uk

Tina F. Keil

apps4research, Pforzheim, Germany
tina.keil@a4r.eu

Mark Levine

Lancaster University, Bailrigg,
Lancaster, UK, LA1 4YW
mark.levine@lancaster.ac.uk

Avelie Stuart

University of Exeter, Exeter, UK, EX4
4PY
avelieresearch@outlook.com

Bashar Nuseibeh

The Open University, Walton Hall,
Milton Keynes, UK, MK7 6AA, Lero,
University of Limerick, Ireland, V94
T9PX
bashar.nuseibeh@open.ac.uk

Arosha K. Bandara

The Open University, Walton Hall,
Milton Keynes, UK, MK7 6AA
aroshabandara@open.ac.uk

ABSTRACT

A plethora of psychological research suggests that gratitude interventions can improve individual well-being and prosocial behavior. However, most of the existing studies focus on gratitude interventions of individuals within controlled lab-based experiments. Our work aims to explore how people practice public gratitude in their everyday lives. This paper presents the design of our public gratitude platform that allows users to freely post gratitude as a tree leaf. Based on usage data spanning 3 years since its launch, our exploratory data analysis provides empirical insights into how people engage in gratitude practices ‘in the wild’. Six types of gratitude, together with their characteristics are identified: personal, work-related groups, study-related groups, gratitude to public service, local community, and event-triggered gratitude. This paper highlights the organic growth of our gratitude platform and its innovative use. Furthermore, the limitations of our work are discussed, shedding light on potential opportunities for future research.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

Gratitude, technology-facilitated gratitude intervention, social gratitude

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1 INTRODUCTION

The ‘Clap for Our Carers’ campaign [9] received wide recognition during the global pandemic COVID-19 [31], where people expressed thanks to key workers by clapping or making noises every week [13]. Gratitude has been defined variously in literature [20, 37, 40]. It generally involves positive emotions in response to benefits received from others [6] or appreciation of various positive aspects of one’s life [3, 37]. A series of psychological studies have demonstrated the positive impact of gratitude on people’s well-being and

health [14, 15, 38, 41]. Personal expression of gratitude may also yield a variety of benefits, including decreased stress levels [42], enhanced social relationships [41], as well as increased pro-social behavior [26]. A workplace culture built on gratitude could increase employee retention, engagement, and productivity [18]. Additionally, gratitude-based interventions, such as a gratitude diary [11, 14] and the “*Three Good Things*” exercise [33], have been introduced in healthcare, e.g., clinical counseling [5].

With a proliferation of gratitude applications (apps) on the mobile app market, technology-facilitated gratitude interventions have also been increasingly explored by human-computer interaction (HCI) researchers. Some researchers have examined the benefits of exposure to other’s gratitude on social media, such as *Facebook* [32], and *Instagram* [24], on the other hand, others have explored new gratitude-based interventions for online communities [29] or via innovative ways of interactions, such as augmented reality [7] and voice control via *Amazon’s Alexa* [21]. However, most existing studies focus on gratitude interventions for individuals, predominantly within controlled experimental settings. In addition, a few mixed effects of gratitude intervention have been noticed by psychologists, including negative socially oriented emotions (e.g., indebtedness and guilt [37]), the pressures of filling a daily diary [21], and personal characteristics (e.g., depression [22], motivation [41]) which may affect self-initiated gratitude interventions. Most importantly, there is limited research on gratitude interventions at the group, community, or societal level. *How people perform public gratitude interventions as part of their lives* is still an open question.

Originally driven by the aim of aiding people to cope with the adversities caused by the COVID-19 pandemic [30], we built and launched a website that allows the public to express gratitude freely and anonymously in December 2020. Since then, it has witnessed remarkable organic steady growth without any external promotions or paid advertisements. Based on a preliminary analysis of three-year usage data (24982 gratitude messages), this study explores how people use the public platform to express gratitude ‘in the wild’. Six types of gratitude are identified with unique characteristics reported. We also discuss the implications of future research in designing public gratitude interventions.

2 PUBLIC GRATITUDE PLATFORM

Inspired by the positive experience [27, 43] and post-traumatic growth [10, 36] found during the COVID-19 pandemic, we designed a public gratitude platform called Gratitude Tree (<https://gratitude-tree.org/>, see Figure 1, left), where users could create, view, and share gratitude both individually and collectively. The platform was evaluated internally among authors and their networks before being launched for general use. We present the primary design considerations based on psychological theories and ethics, followed by an overview of the platform’s functionalities.

2.1 Design Considerations

2.1.1 Why Public Gratitude. Although most research has focused on individual-based gratitude, there is a growing interest in exploring gratitude intervention at the group or social level, such as group gratitude [13] and collective gratitude [28]. It demonstrates that the group context, where individuals are aware of others’ practice of

gratitude, can encourage a “*cycle of gratitude*” and other prosocial behavior [25]. In other words, people in the group might practice more gratitude and acts of kindness in return. For example, Di Fabio et al. [18] found that organizational gratitude strengthens employees’ productivity and success, also improves organizational citizenship and prosocial organizational behavior. Another study shows that exposure to grateful posts or comments on *Facebook* could improve subjective well-being [32], which correlates with spreading positivity in online social ecosystems [19].

Informed by the theory of social learning [4] and emotional contagion [32], public disclosure of gratitude might promote gratitude as a collective virtue. Our platform creates an online space for reflection and cultivating a positive environment: users could get inspired to reflect on positive aspects of their own lives by browsing other’s gratitude messages. Users are also able to interact with others’ gratitude by adding more gratitude to the public space. The platform could also be used by organizations and local communities to promote their gratitude initiatives and enhance a culture of gratitude. Additionally, we follow the ethics guide by Townsend and Wallace [35], especially related to public data and anonymity. Hence, our platform doesn’t require account registration, and it does not track any user information (e.g., cookies). To give users the freedom to edit or remove their contributions, an authentication system for modification purposes has been added to our platform. Alternatively, users could contact us to withdraw any contributions.

2.1.2 Tree and Forest Metaphor. Metaphors are powerful narrative and cognitive devices that enhance the clarity of complex concepts and evoke emotions [39]. Trees are widely used as metaphors for broad concepts like life, history, health, and society [39]. For example, tree-like diagrams have been used in daily contexts to illustrate complicated relationships like a family tree. Leveraging a familiar metaphor like a tree provides a more engaging and user-friendly interface and a consistent user experience.

The metaphor of the tree and forest effectively conveys the essence of public gratitude and its benefits. The gratitude messages are attached to “*leaves*” on the “*trees*” in the “*forest*”. People also could “*plant*” an empty “*gratitude tree*” to encourage others to add leaves of gratitude. Our platform symbolized as a “*forest of thanks*”, simulates the idea of a communal space where individuals can share gratitude, cultivating a collective growth of positivity like the growth of a tree in the forest. Just as trees spread their seeds to reproduce, a single leaf of gratitude may inspire other users of our platform to plant more trees or leaves of gratitude. The different shapes of trees (see Figure 1, right) and color choices for a leaf reflect the diverse nature of the ecosystem in the forest [16]. The ability to modify trees and leaves represents a living dynamic ecosystem that can continuously “*morph*” its species. The “*foresters*” are the administrators of the website, who monitor and “*prune*” trees or leaves that become “*diseased*” with inappropriate messages.

2.2 Main Functionalities

The main functionalities of our platform include creating/modifying a gratitude tree, creating/modifying an individual gratitude leaf, searching a gratitude tree/leaf, viewing/sharing a gratitude tree/leaf, subscribing to a tree for updates, and downloading a tree with all gratitude messages on it.

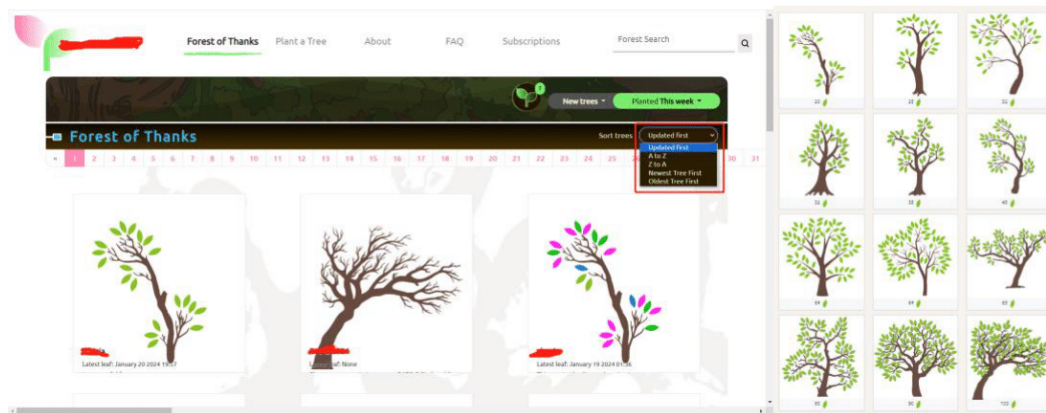


Figure 1: Left: The screenshot of our platform’s “Forest of Thanks”. Right: Pre-defined shapes of trees with various amounts of leaves (20, 25, 32, 35, 45, 64, 65, 80, 90 and 100).

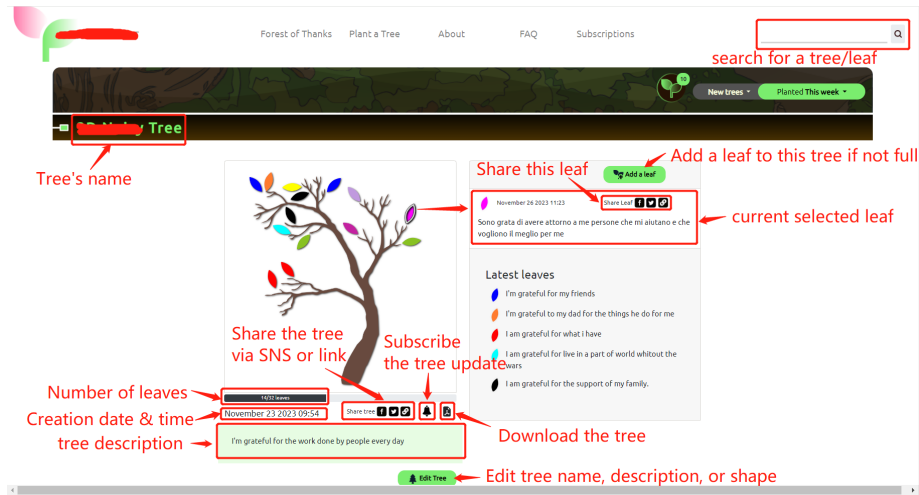


Figure 2: The screenshot of the tree view, annotated with functions of viewing, sharing, modifying, subscribing, adding a leaf, and downloading a tree.

- **“Plant a tree”**: A user can create a gratitude tree with a name and a tree shape with different amounts of leaves (see Figure 1, right), and the password could be set up optionally if a user wants to modify the name, description, and shape of the tree later.
- **“Forest of Thanks”**: All gratitude trees are openly accessible to the public in the forest of thanks (see Figure 1, left), where a user can browse the trees by sorting trees based on updated trees first, newest tree first, oldest tree first, and alphabetically by the tree name (A to Z, or Z to A).
- **“Add a leaf”**: Users can add a leaf with any color to any gratitude tree that has room to add new leaves. A textbox with a limit of 5000 characters enables users to input free text or a Tweet URL. A password could be added optionally

to the new leaf to allow later modification of the gratitude message and color of the leaf.

- **Modify a tree or leaf**: After verifying the original password of a tree or a leaf, the name, description, and shape of the tree could be modified; or the message and the color of the leaf can be modified.
- **Browse a tree or leaf**: By default, there are five most recent leaves displayed for each tree, with the tree name, description, creation time, and the number of leaves left for new leaves (see Figure 2). Users can click any leaf to view, share, and modify the gratitude message.
- **Sharing a tree or leaf**: The tree view also enables a user to share the tree or a leaf via social media (*Facebook* and *Twitter*) or by copying the link of the tree or the leaf (see Figure 2).

- *Subscribe to the tree*: a notification will be sent to the user's browser when the subscribed tree has a new leaf. A user could remove the subscriptions via the 'Subscriptions' page of the site.
- *Search for a tree or leaf*: when searching by a keyword, all trees with the tree description or a leaf related to the search keyword will appear.

3 METHODOLOGY

This study focuses on the preliminary exploration of how users use trees and leaves to express gratitude in a real-world context. After the initial launch in December 2020 and the authors' social media promotions in the initial five months, the website has experienced consistent growth. Following the ethical guide for social media research [35], we conducted an exploratory analysis using data from 1679 trees and 24982 leaves created between 14th December 2020 and 3rd January 2024. We collected the following information from our website:

- For a gratitude tree: the creation time of the tree, the creation time of the last leaf, the tree's name, description, the number of leaves, the tree's shape, and the password status (yes/no).
- For a gratitude leaf: the creation time of the leaf, the password status (yes/no), and the gratitude message.

Descriptive statistics were produced using *Microsoft Excel*. The "growing period" of the tree (the time elapsed from its creation to the last leaf being added) and the word count of both the tree description and gratitude leaf were also analyzed. Two researchers analyzed tree descriptions and the gratitude messages on each leaf based on an agreed codebook for tree categories (based on user purposes) through an iterative process. Weekly discussions between researchers during the analysis phase enhanced the validity of the data analysis.

4 FINDINGS

Findings indicate that our platform has been steadily growing since its launch (see Figure 3). Six categories of gratitude were identified based on inferred purposes for creating trees and adding leaves: personal, work-related group, study group, gratitude to public service, location-based community, and event-triggered gratitude. We also found examples of users appropriating our platform for other purposes. We report the characteristics of each type of gratitude tree including the length of the gratitude message and active status.

4.1 Six Categories of Gratitude Tree

4.1.1 Personal Gratitude. Personal gratitude trees are created by individuals to reflect and express gratitude for the general positive aspects of their lives, such as family, friends, and things or people offering support and help. Personal gratitude messages tend to be the **shortest** among all gratitude categories, with an average length of 4.9 words. Most personal gratitude trees (85.1%) grew their leaves within one day. Users tended to not provide descriptions for their personal gratitude trees, and only 19.1% of trees have descriptions, which are typically short (on average 8.5 words). Additionally, only a few users specified a password for their trees (29.4%), suggesting

that most do not intend to modify trees and reinforcing the observation that people's engagement with personal gratitude trees has the **shortest duration of being active** (averaging 6 days).

4.1.2 Work-related Group Gratitude. Gratitude trees within work-related groups were typically initiated by an employee or a manager to express gratitude directly to colleagues or to encourage all employees to practice gratitude. Most work-related group gratitude trees have a medium-length description (70.1%), averaging 20.9 words. Notably, 59.1% of work-related group gratitude trees have a password, indicating users are willing to modify trees, with an active period of 19 days. Each gratitude message is **relatively long** with completed sentences, averaging 15 words.

4.1.3 Study Group Gratitude. The study group gratitude trees were used by school or university students, or clients of advisory groups. We found that 65.8% of study group gratitude trees have the **longest tree description** (averaging 30.3 words) to guide or encourage others on the purpose of the tree and how to add leaves of gratitude. More than half of these trees have a password (55.3%), and users interact with a tree for an average of 11 days. The **longest narratives** are used for expressing gratitude within the study group, averaging 15.9 words.

4.1.4 Gratitude to Public Services. Public service-related gratitude includes the benefit receivers' appreciation of healthcare providers, funders, religion, and school. It is the **most active** type of gratitude tree, with an average period of 40 days for each tree. This means that users keep adding gratitude messages to the public service tree over a longer period. Most public service gratitude trees have a tree description (79.3%, averaging 26.6 words) and a password (66.7%).

4.1.5 Location-based Community Gratitude. Location-based community gratitude trees were created for the local community to jointly share their appreciation. The gratitude messages are relatively **short** (on average 8.7 words), but each tree typically has **more leaves** than other types of gratitude trees (32.3 leaves on average), and it tends to have a password (66.7%) and tree description (66.7%, averaging 17.5 words), with averaging 21 days of being active.

4.1.6 Event-triggered Gratitude. The platform has the highest utilization during special occasions including festivals and events like Thanksgiving Day. The event-triggered gratitude trees were used to express gratitude towards individuals, public services, or within work-related or study groups. Most event-triggered gratitude trees have a description (71.4%, on average 22.3 words) and a password (61.5%). The active period of an event-triggered gratitude tree tends to be relatively **short** (8 days), but each gratitude message has 14.7 words on average.

4.1.7 Testing Tree and Creative Use of Our Platform. Users attempted to explore how to use our platform by creating a test tree, especially for organizational use. Most testing trees are interacted with once (88.9%) and have no tree description (66.7%). People created the shortest leaf (on average 1.9 words), sometimes without meaningful content.

Furthermore, it has been observed that study-related groups are using our platform in innovative and unforeseen ways. For example, a school used our platform as a visualization tool to present

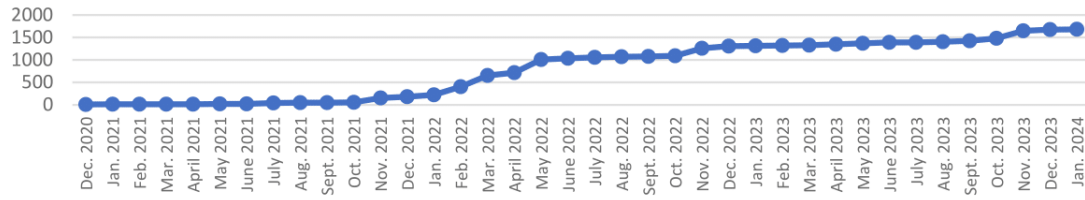


Figure 3: The cumulative total number of trees each month.

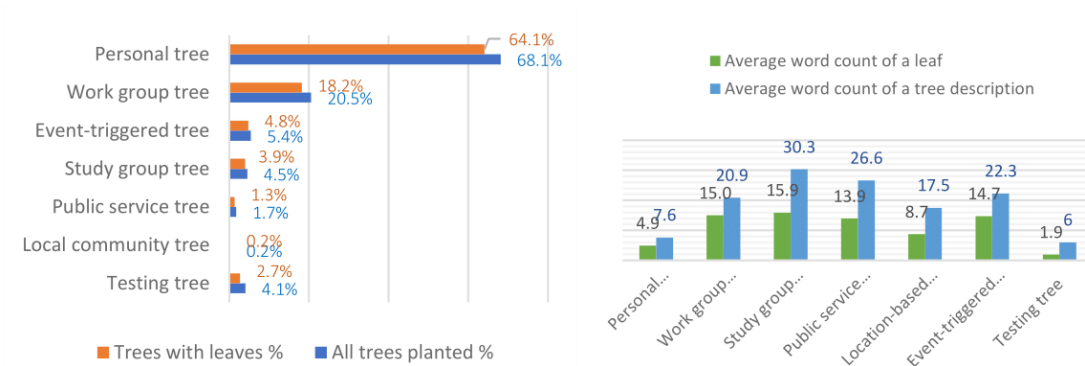


Figure 4: Left: Percentage distribution of gratitude tree categories; Right: The average word count of a leaf and a tree description.

teachers' information. Other schools have adopted our platform to facilitate and encourage their students to pledge commitments to pro-environmental activities, designating a tree as a collective pledge from one class.

4.2 Overall Use

The predominant use of our public gratitude platform is for personal purposes (68.1%, see Figure 4, left), followed by work-related groups (20.5%), event-triggered gratitude (5.4%), and study groups (4.5%). Location-based community gratitude involved the least engagement, accounting for only 0.2%. Most gratitude trees represent one-time interactions (73.6%), and 9% of all gratitude trees had no accompanying leaves of gratitude. Passwords are not used often: only 38% of gratitude trees and 7.3% of leaves were created with a password for modification purposes. Half of the gratitude trees (52.4%) have a meaningful description that is not garbled text or emojis. The average word count for a tree description and a gratitude message can be seen in Figure 4 (right). Our initial findings also show that multiple languages are used, such as Spanish, Thai, and Arabic, indicating that users may come from different countries and cultures.

5 DISCUSSION AND FUTURE WORK

The exploratory data analysis shows the organic growth and diverse use of our public gratitude disclosure platform within a broader social context. This suggests people's general need to practice

gratitude at both individual and group levels over time. The ecological usage observed on our platform aligns with Cockayne and Salter's taxonomy of group gratitude [13], highlighting the complex and multifaceted nature of gratitude. This study demonstrates the diverse expression of gratitude in the real-world context, which reflects various conceptualizations of gratitude documented in prior research [13, 17, 37] – referring to gratitude as an emotion, a personality trait, an action, or a moral virtue. Our findings echo the seminal work of Gulliford et al. [20], urging researchers to further examine fundamental concepts of gratitude.

This study offers empirical insights into various gratitude categories and their features. The platform itself could be perceived as a living lab [1], where the participants collectively answer the questions: *what are you grateful for?* In contrast to traditional lab-controlled experiments, the usage pattern observed on our platform provides researchers with a comprehensive view of how the public engages in gratitude expressions in real-world settings. This insight is important, especially for those in the field of positive computing [8] who strive to design digital technologies that support and flourish human well-being.

Findings indicate that gratitude for personal use is the most popular among the six categories of gratitude, with the gratitude message being the shortest with an incomplete sentence. It is not surprising that gratitude within a work or study group has the longest words because employees or students might feel obligated to use the completed sentences to articulate their thanks or follow

the norm of reciprocity [37]. It is reasonable to observe that gratitude trees for public service have the longest active period, where a broader group of users respond to benefits identified at the collective level [13]. It is worth noting that six categories of gratitude are not exclusive. For example, the leaves of the personal gratitude tree could include the expression of gratitude toward an individual's work or study. It suggests the complexities of each gratitude category, which opens opportunities for further exploration, such as the correlation between different gratitude categories. Exploring gratitude themes over different periods (e.g., during the pandemic vs. post-pandemic) would also help in understanding factors of people engaging with self-initiated gratitude practice.

Limitations & Future Work: The nature of our public gratitude platform introduces some limitations. Firstly, users self-initiate their discovery and engagement with our website to practice gratitude, so it does not represent the whole population with all personal characteristics [34]. Secondly, due to the anonymity of our platform, there is a lack of details about participants, e.g., the quantity and geographic distribution of users. However, the anonymous usage of our platform may encourage users to share authentic narratives freely. Most importantly, this approach values the care and ethical considerations for a technology-based gratitude intervention.

Future work could conduct a qualitative analysis of gratitude leaves and a further study to explore the effects of using our public gratitude platform. An essential aspect could be understanding the reasons behind the short period of engagement, where people may not add gratitude messages after the first visit to our website. This also raises a pivotal question: should researchers or designers prioritize strategies for sustainable engagement (e.g., gamification [12], prompt [23]) in technology-facilitated gratitude interventions? Moreover, it suggests critically examining the effects of using a public gratitude platform 'in the wild', especially potential harms or burdens [40], e.g., indebtedness [37] or manipulation [40], as cautioned by a few researchers. Exploring the link between gratitude practices and acts of kindness [2] is also a promising work.

6 CONCLUSION

This work explores how people practice public gratitude in the real-world context. We built a public gratitude platform by using the metaphor of a tree and forest. Based on data spanning three years of gratitude practices, our preliminary exploratory data analysis identifies six categories of gratitude and their characteristics. The overall usage patterns and organic growth of our platform are also highlighted. This work contributes to empirical insights into the ecological use of public gratitude. We discuss both the findings and limitations of this work, highlighting the opportunities for future research.

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